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# Artem Makoian

Languages: Python3, C++, C, Go  
github.com/MakArtKar  
codeforces.com/profile/MakArtKar  
leetcode.com/MakArtKar

## SKILLS

**Libraries** Pytorch, Transformers, Hydra, DVC, WandB, OpenCV, Sklearn, Numpy, Pandas, ONNX, Seaborn  
**Tools** SQL, Linux, Bash, Git,  $\LaTeX$ , Markdown

## TECHNICAL EXPERIENCE

 **JetBrains x Constructor University of Bremen (PhD Student)** Feb 2025 — Present  
ML Researcher / JetBrains Research with D. Vetrov Python 3, Pytorch, transformers, Hydra, Lightning

- Latent Diffusion Drafters for Lossless Acceleration of LLM Inference
- Distillation (flow maps, consistency) and latent space compression for real-time acceleration.
- Guided Star-Shaped Masked Diffusion - ICML 2026: main track; ICLR 2026: ReALM-GEN Workshop
- Dream7B fine-tuning to enable self-correctness mechanism - already generated tokens improvement

 **Yandex** Jun 2023 — Sep 2024  
Middle ML Engineer/ Geo services Python 3, LangChain, PyTorch, Hydra, OpenCV, YT, Shapely, YQL, Nirvana

- Map roads segmentation from aerophotos with IoU 0.93
- Roads post-processing algorithms - reduced number of artifacts from 10% → 0.02% and improved visual quality.
  - Safety islands segmentation and vectorization with IoU 0.37
- News classification to create map updates hypotheses. Precision: 0.9, recall: 0.85.
- Developing and maintenance of the news pipeline: site parsing (with LangChain AI agents), structuring data and classifying.

 **Sber Devices** Jul 2022 — Jun 2023  
Middle ML Engineer/ Computer Vision Platform Python 3, C++, PyTorch, Hydra, DVC, OpenCV, ONNX, torch\_pruning

- Background removal for video conferencing with 0.98 dice metric
- Improved human segmentation quality working on segmentation pipeline, pruned model getting 1.5x speed acceleration
  - Improved visual quality with post-processing algorithms: aligning the lighting and smoothing out the border.
- Trained 700KB model for pay card border segmentation and developed C++ inference pipeline in multiplatform mobile SDK

 **Yandex** Jul 2021 — Oct 2021  
ML Developer intern / VR Team with V. Lempitsky Python 3, PyTorch, Tensorboard

- Trained model that predicts human's alpha mask to improve the quality of a VR scene
- Prepared datasets with human alpha masks, read and compared papers on an alpha matting problem
  - Trained an improved FBA Matting net without using trimaps

 **Samsung R&D Department** Nov 2020 — June 2021  
ML Research Intern / Mobile Authentication Python 3, PyTorch, Bash, Numpy, Pandas, Sklearn, Tslearn

- Worked on a pipeline for mobile phone authentication by user's motion patterns with  $82.2\% \pm 9\%$  accuracy - paper.
- Collected, preprocessed and handled data from devices. Used ML and DL models for authentication, got 70% accuracy baseline

## EDUCATION

 **Master in Computer Science, Constructor University of Bremen, Germany** Sep 2024 — May 2026

 **Bachelor in Computer Science, GPA 86/100, Neapolis University of Paphos, Cyprus** Jan 2023 — May 2024

 **Bachelor in Computer Science, GPA 8.7/10, Higher School of Economics, Moscow, Russia** Sep 2019 — Jun 2024

 **Master's Equivalent, Yandex School of Data Analysis, Data Science, Russia** Sep 2020 — May 2022

### Relevant courses:

- Deep Vision, CV, NLP, 3D CV, Generative Models, RL, Bayes Methods, Optimization Methods, Deep Bayes, Large Scale DL
- ML, DL, Python, C++, Go, Algorithms and Data Structures, Matrix Calculations, Statistics, Linear algebra, Probability theory

## PROJECTS

**Telegram RAG QA bot** Jun 2023 — Aug 2023

- Developed tg bot to answer on questions giving it the chat context Python 3, LangChain, HuggingFace, aiogram
- Researched different LLM models, developed different types of LangChain pipeline (stuff, rerank, map reduce, refine) with them.

**Automatic Detection of Means of Transportation on 3d Cargo Models** Dec 2021 — June 2022  
Python 3, Pytorch, PyTorch3D

- Developed model for cloud segmentation with 0.95 IoU for cargo
- Calibrated cameras for cargo scanning, prepared data, implemented ESANet model

## COMPETITIONS

 Awards in All-Russian Olympiad: 33<sup>rd</sup> place in **Informatics** and 8<sup>th</sup> place in **Maths** over 20000 participants Apr 2018, 2019

 **Top 8%** in Google Hash Code. **Top 50** in Google Kick Start. **Top 350 (top 0.8%)** in Google Code Jam Feb, May, June 2020